

Superconducting nanowires at the ePIC far-forward: what a closer approach would buy the polarized-lithium tags

2026-08-26 · companion to plans/09 and to open questions #19 and #20 · computed with

`evgen/scripts/nearbeam_aperture_scan.py`, `nearbeam_reach_gain.py` and `nearbeam_sensor_budget.py` on `polligen.reco`, `polligen.recopseudo` and `polli_fastsim.spectator` · primary sources in `refs/`

The far-forward tags of the polarized-lithium programme are beam-blind. An intact ${}^6\text{Li}$ has $A/Z = 2$ exactly and the α from its breakup has 0.99813 of the beam's rigidity, so neither is separated from the beam by dispersion: what separates them is the transverse *angle* at the interaction point. The acceptance is therefore $\exp(-B|t|)$ evaluated at $|t| = (\theta p)^2$ — an exponential in the *square* of the approach angle times the square of the momentum — and approach distance is the whole measurement. Superconducting nanowire detectors, developed for nuclear physics by the Argonne MEP and Physics Divisions, are the candidate technology. This note prices what a closer approach is worth, asks whether a nanowire can deliver it, and separates the part of the case that survives scrutiny from the part that does not.

The verdict, and it is negative. Neither proposed role survives, and both fail on arithmetic rather than on unknowns. (1) **The near-beam role.** A closer approach is worth a great deal — $\times 26$ at 5×41 , $\times 569$ at 10×100 , and at mid energy the difference between a fit that returns $a_t = 0.48 \pm 1.45$ and one that returns four clean $|t|$ bins (§2). But the gap is set by 16 mm module tiling and by the 10σ rule, and a nanowire's advantage over slim-edge silicon is its dead edge — worth 0.005 mrad against a $0.5\text{--}3 \text{ mrad}$ gap. The 10σ rule is machine protection and halo, not sensor survival, so radiation hardness does not relax it either (§3). (2) **The charge-ID role.** A nanowire latches and delivers *one bit* per plane — and that turns out to be *almost enough*: against the Neyman-Pearson optimum a one-bit, four-plane threshold loses a factor 1.4, while the standard analogue estimator (a truncated mean) loses a factor 86 to one bit. Bits are not the issue. What kills the nanowire is **geometric fill factor**: the coincidence that makes one bit sufficient needs every plane to record the track, and at the ANL device's 50% fill a four-plane array cannot reach 95% ${}^6\text{Li}$ efficiency at any working point (§4.1). (3) **Open question #19 has been asked of the wrong technology.** Better still, the $\alpha + d$ breakup that #19 exists to reject is *two hits*, and at beam rigidity they land 6.7 to 45 mm apart — 13 to 90 pixels of the existing $500 \mu\text{m}$ pitch. Topology, not dE/dx (§4.3).

What does survive, and it is the reason to read on. The acceptance-versus-aperture curve of Figure 1 is technology-independent: it prices *any* closer approach, by any means, and says the mid-energy configuration is worth $\times 569$. And a correction that is independent of nanowires and more immediately actionable than anything else here — **the ePIC pot geometry has moved since the snapshot this repository measured**, in opposite directions at the two ends (§6). Every aperture-conditional number in the programme needs re-measuring.

Scope note (2026-08-28). The tagged fractions and the "10 σ envelope" of this report are conditioned on the proton-derived, energy-independent, isotropic divergence $\sigma_\theta = 72.7 \mu\text{rad}$ ($10\sigma = 0.727 \text{ mrad}$) that every number before 2026-08-27 used. The Yellow Report's own tables give 220/380, 180/180 and 92/92 μrad (h/v) for ${}^6\text{Li}$ at the three configurations (plans/10, Report 3 §4.1), under which the envelope alone leaves no tag at any of them and the near-beam gains below shrink to the ratios they are; the reconstructed-level coherent chain has since been re-derived at the lithium tagging optics of Report 1 §6.1 (Report 2 §4.5, §5.2). The mechanism arguments of §§1–5 stand; the absolute acceptances of Figure 1 and Table 1 are superseded (plans/10 A4).

1 · Why the angle is everything

Both far-forward lithium tags are rigidity-degenerate with the beam. The coherent measurement needs the intact ${}^6\text{Li}$, whose A/Z is exactly the beam's; the tagged measurements need the α spectator, whose rigidity is 0.99813 of it. Neither is swept aside by the dipoles. The only handle is the transverse angle, and because the coherent recoil follows $\exp(-B|t|)$ with $B \approx 50 \text{ GeV}^{-2}$ and $|t| = (\theta_x p)^2$, the tagged fraction falls as the exponential of the *square* of the approach angle times the *square* of the beam momentum. Halving the approach angle at 10×100 is not worth a factor of two; it is worth a factor of 569.

Two angles compete to set that approach. The **10 σ envelope** — 0.727 mrad for the high-acceptance optics — is an operational retraction, a machine-protection rule about beam halo. The **geometric aperture** of the sensor package is what the silicon, its RF shield, its ASIC and its module granularity actually leave open. Shooting an intact ${}^6\text{Li}$ through the ePIC geometry (`tools/fullsim`) measured the second: $|\theta_x| \gtrsim 2.00 / 1.35 / 1.03$ mrad at $5 \times 41 / 10 \times 100 / 18 \times 275$, or $27.5 / 18.6 / 14.2 \sigma_\theta$ — a package that stops 4.2σ to 17.5σ short of where the machine would allow it. §6 revisits whether that measurement still describes today's geometry; the curves below do not depend on the answer.

2 · What a closer approach is worth

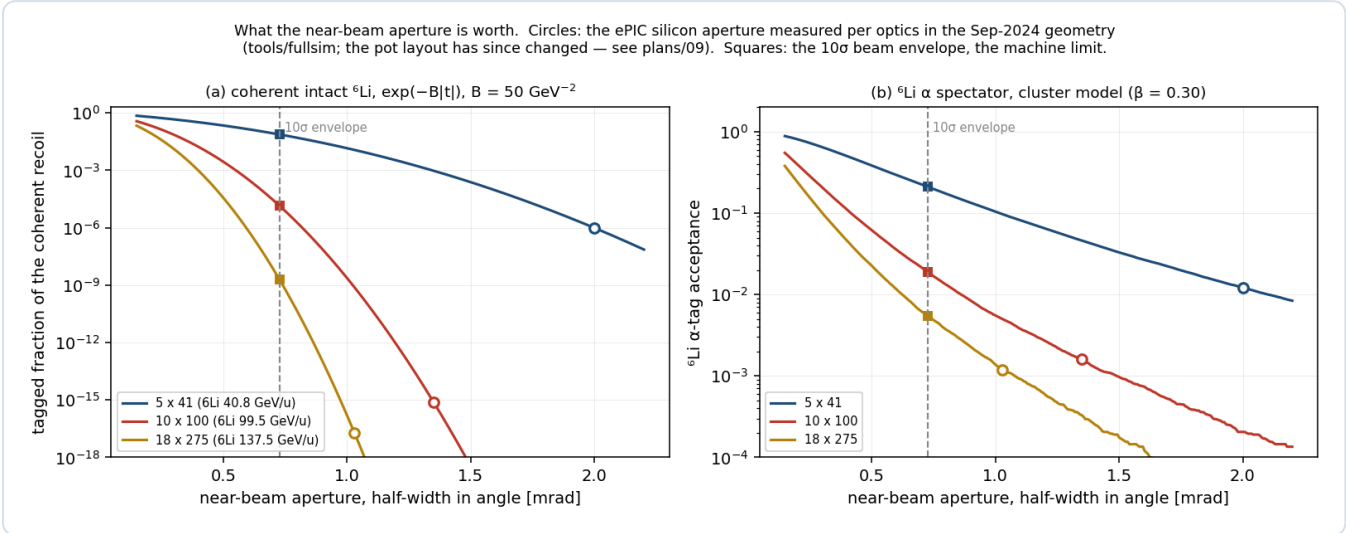


Figure 1 · The price of every aperture. Tagged fraction of the coherent intact ${}^6\text{Li}$ (a) and of the ${}^6\text{Li}$ α spectator (b) against the near-beam half-width in angle, per beam configuration. Circles mark the ePIC silicon aperture measured in the September-2024 geometry; squares mark the 10 σ envelope. The vertical half-width is held at its measured value per optics (3.0 / 3.0 / 2.3 mrad). Note the vertical scale of (a): eighteen decades. `nearbeam_aperture_scan.py` .

The analytic gains are large and very unequal:

	5 × 41 (40.8 GEV/U)	10 × 100 (50)	18 × 275 (137.5)
coherent tagged fraction, silicon aperture	1.41×10 ⁻²	5.12×10 ⁻⁵	1.94×10 ⁻¹⁷
coherent tagged fraction, 10 σ envelope	3.71×10 ⁻¹	2.91×10 ⁻²	1.97×10 ⁻⁹
gain	×26	×569	×1.0×10 ⁸ , still dead
${}^6\text{Li}$ α tag, silicon aperture	0.103	0.024	0.0012
${}^6\text{Li}$ α tag, 10 σ envelope	0.550	0.137	0.0054

Coherent: `reco.rp_hole_acceptance` at $B = 50 \text{ GeV}^{-2}$ with the measured rectangular cutout. α tag: the cluster model of `polli_fastsim.spectator` at $\beta = 0.30$.

The top energy is not rescued, by any technology. At 825 GeV even a 10 σ approach demands $p_T > 0.60 \text{ GeV}$, $|t| > 0.36 \text{ GeV}^2$, and $\exp(-50 \times 0.36) = 10^{-9}$. No aperture makes the coherent channel live at 18×275 ; the client there is the α tag and nothing else.

2.1 · Through the chain, not just the acceptance

Acceptance is not the measurement. Running the full coherent chain — Roman-Pot emulation, the two azimuths $\alpha = \varphi_e - \varphi_S$ and $\beta = \varphi_t - \varphi_S$, the spin-state-sorted 2-D harmonic fit — at both apertures shows three effects, not one.

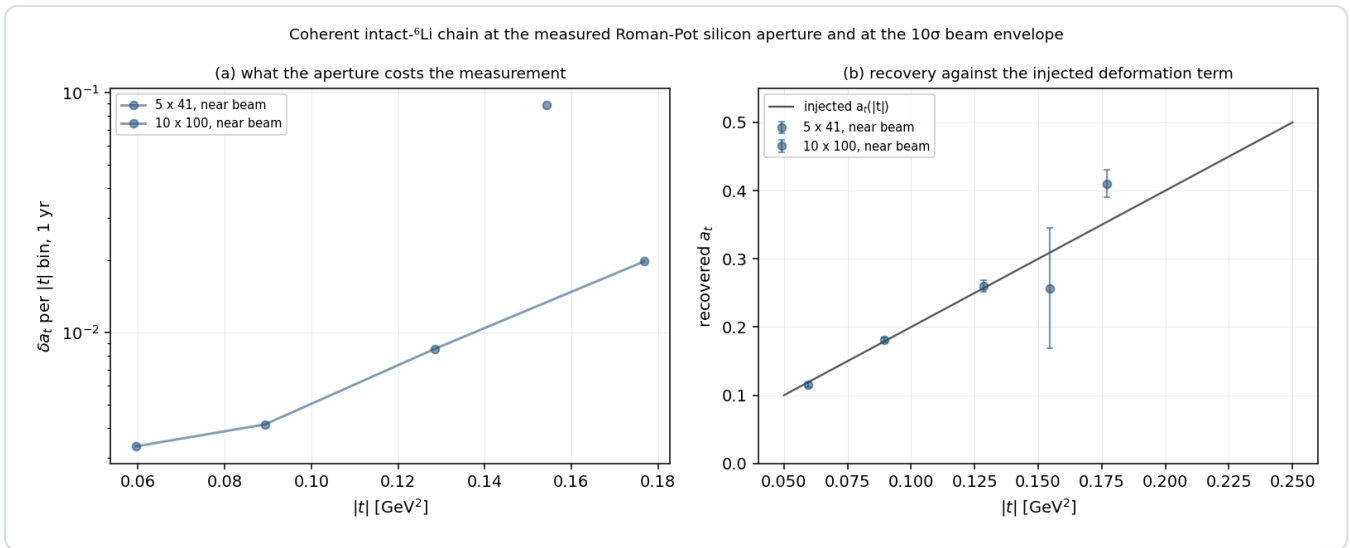


Figure 2 · What the aperture costs the measurement. (a) δa_t per $|t|$ bin at 1 year, at the measured silicon aperture (dashed, squares) and at the 10 σ envelope (solid, circles), for the two configurations where the coherent channel is alive. (b) The recovered a_t against the injected deformation term; the off-scale silicon point at 10 $\times$ 100 is labelled.

`nearbeam_reach_gain.py`.

OPTICS	APERTURE	ACCEPTANCE	$ t $ BINS	ΔA_T PER BIN, 1 YR	A_E (INJECTED 0.0100)
5 $\times$ 41	2.00 mrad, silicon	1.43×10^{-2}	2 of 4	0.046, 0.015	-0.0060 ± 0.0045 , $+0.0197 \pm 0.0044$
5 $\times$ 41	0.727 mrad, 10 σ	3.62×10^{-1}	3 of 4	0.0019, 0.0035, 0.0078	$+0.0084 \pm 0.0016$, $+0.0087 \pm 0.0031$, $+0.0012 \pm 0.0076$
10 $\times$ 100	1.35 mrad, silicon	1.02×10^{-4}	2 of 4	1.45, 0.175	-0.068 ± 0.049 , -0.019 ± 0.020
10 $\times$ 100	0.727 mrad, 10 σ	3.25×10^{-2}	4 of 4	0.0069, 0.0044, 0.0077, 0.0168	$+0.0117 \pm 0.0013$... $+0.0304 \pm 0.0118$

- Statistics.** δa_t improves $\times 8$ to $\times 40$ where the measurement already exists.
- A configuration.** At 10 $\times$ 100 the silicon aperture returns $a_t = 0.48 \pm 1.45$ against a truth of 0.32 — not a measurement. At 10 σ it returns four clean bins. The mid-energy configuration goes from dead to the best-covered in the programme, which matters because the reconstruction-chain report otherwise concludes that the coherent channel is a low-energy-only programme. *That conclusion is aperture-conditional, not physics-conditional.*
- Bias, not only error.** At 5 $\times$ 41 with the silicon aperture the gluon-transversity coefficient comes back -0.0060 ± 0.0045 and $+0.0197 \pm 0.0044$ against an injected 0.0100 — both wrong by more than 2σ , because the emptied azimuthal circle leaves the seven harmonic columns nearly degenerate. At 10 σ every bin is consistent. A tight aperture does not merely cost events; it biases the extraction.

A trap this exposed. Money plot 6R carries `cut_scale_x = 2.5` from the pre-measurement belief that the pots surround a *wide horizontal slot*. With the geometric aperture measured, carrying 2.5 as well imposes a 25 σ horizontal retraction that no machine requirement asks for — and it binds *before* the geometry does, hiding the entire effect. Both new scripts hold the envelope at 10 σ in each axis and let the measured geometry be the only aperture.

3 · The idea is already on record

Near-beam recoil tagging is not a new application to propose to the Argonne group. It is theirs, twice over. The EIC Yellow Report §14.5.3 lists the four SNSPD applications the EIC R&D committee identified, of which the first is "*a Roman pot detector in the forward region about 35 meters or more from the interaction point to tag low momentum transfer recoiling ions*" [1]. Armstrong's FY22 EIC generic-detector-R&D proposal — reviewed and funded — states the mechanism

verbatim: "The so-called 10 σ rule-of-thumb for Roman pots prevents the beam from damaging the detectors by limiting its proximity to the beam. Radiation hard nanowire detectors will be able to move closer to the beam, thus extending the acceptance reach at lower t " [2].

What this study adds is the number attached to *lower t*, for a species whose acceptance is exponentially more sensitive to it than the DVCS protons that motivate the rest of the far-forward programme. Table 2 is that number.

But the stated mechanism does not hold. The 10 σ rule is a machine-protection and beam-halo rule, not a statement about sensor damage: it exists to keep the *beam* safe and the detector out of the halo, and a more radiation-hard sensor does not relax it. LHC ran the CT-PPS pots at 25 σ at 6.5 TeV. The ePIC geometry says as much in the file itself — the per-energy insertions are "the ten-sigma cuts for the Roman pots, translated to the physical layout we currently have" [3].

And the fallback argument does not survive arithmetic either. One could argue granularity and dead edge instead: the insertion is quantised by the module, and a nanowire's active area reaches within a few hundred nanometres of the substrate edge [1] — measured, in the SMSPD case, to be sharper than the 10 μm telescope used to look at it [8]. That is a real and unusual sensor property. It is simply not what limits this measurement: slim-edge planar and 3D silicon already deployed in Roman Pots reaches 100–200 μm of dead edge, so the difference is $\approx 150 \mu\text{m}$, or **0.005 mrad at $R_{12} = 30.6 \text{ m}$** — 1% of the 0.52 mrad edge and 0.3% of the 10×100 gap. Unmeasurable in $|t|$.

The granularity half of the argument does survive — and it belongs to ePIC, not to a detector R&D group. At 10×100 the blind block is 32 mm wide against a $10\sigma_x$ of order 10–17 mm: a factor ~ 330 in coherent yield surrendered to retracting a whole 32 mm-wide block. The fix is a staircase retraction near $x = 0$, narrower modules, or the x -moving layer ePIC is *already designing* — Jentsch, July 2025: "Simplest option would be one 'x' moving layer and one 'y' moving layer in each station ... Expect to have this finalized in the next 6-8 weeks." This is the highest-value item in the study, and it is a layout change to a baselined detector, not new technology.

4 • Charge identification: the mechanism is right, the technology is not

Open question #19 has no answer at IP6. An intact ${}^6\text{Li}$, an α and a d from breakup share rigidity *and* velocity; only dE/dx separates them, going as z^2 — 9 : 4 : 1 — and the EICROC AC-LGAD chain records pulse amplitude for charge-sharing but no EIC document addresses Z-ID in the pots. The only documented concept is a z^2 Cherenkov behind the IR-8 secondary-focus pots.

A nanowire will not do it by pulse height. The device latches: the pulse amplitude is the diverted *bias* current, and both Fermilab/JPL beam papers show the amplitude distributions from 120 GeV hadrons, 120 GeV muons, 8 GeV pions and 8 GeV showering electrons to be indistinguishable, with one amplitude threshold serving all [7,8]. Any scheme that reads Z off a pulse height is dead on arrival.

It would do it by firing threshold. In the normal-core hot-spot regime a deposit Q makes a normal core of radius $r_s \propto \sqrt{Q}$, and the wire switches when that core spans it against the bias-suppressed superconducting width:

$$r_s = \sqrt{\frac{Q}{e\pi c\rho(T_c - T_0)}}, \quad \frac{I_{th}}{I_c} = 1 - \frac{2r_s}{w}$$

Three groups converge on this relation from three directions — Argonne's own Eqs. 1–2 for 120 GeV protons [4], Renema's $E = (w/C)^2(1 - I_b/I_c)^2$ for photons, and the ion literature's $I_{th}/I_c = 1 - (zeV)^{1/2}C/w$ [9]. Because $dE/dx \propto z^2$ at fixed velocity, and because a ${}^6\text{Li}$ at 137.5 GeV/u has $\beta\gamma = 148$ against 128 for Argonne's calibration proton — the same velocity to 15% — **r_s scales as z linearly, with no velocity correction:** 134 nm (p, d), 268 nm (α), 402 nm (${}^6\text{Li}$), anchored on Argonne's own measured 134 nm.

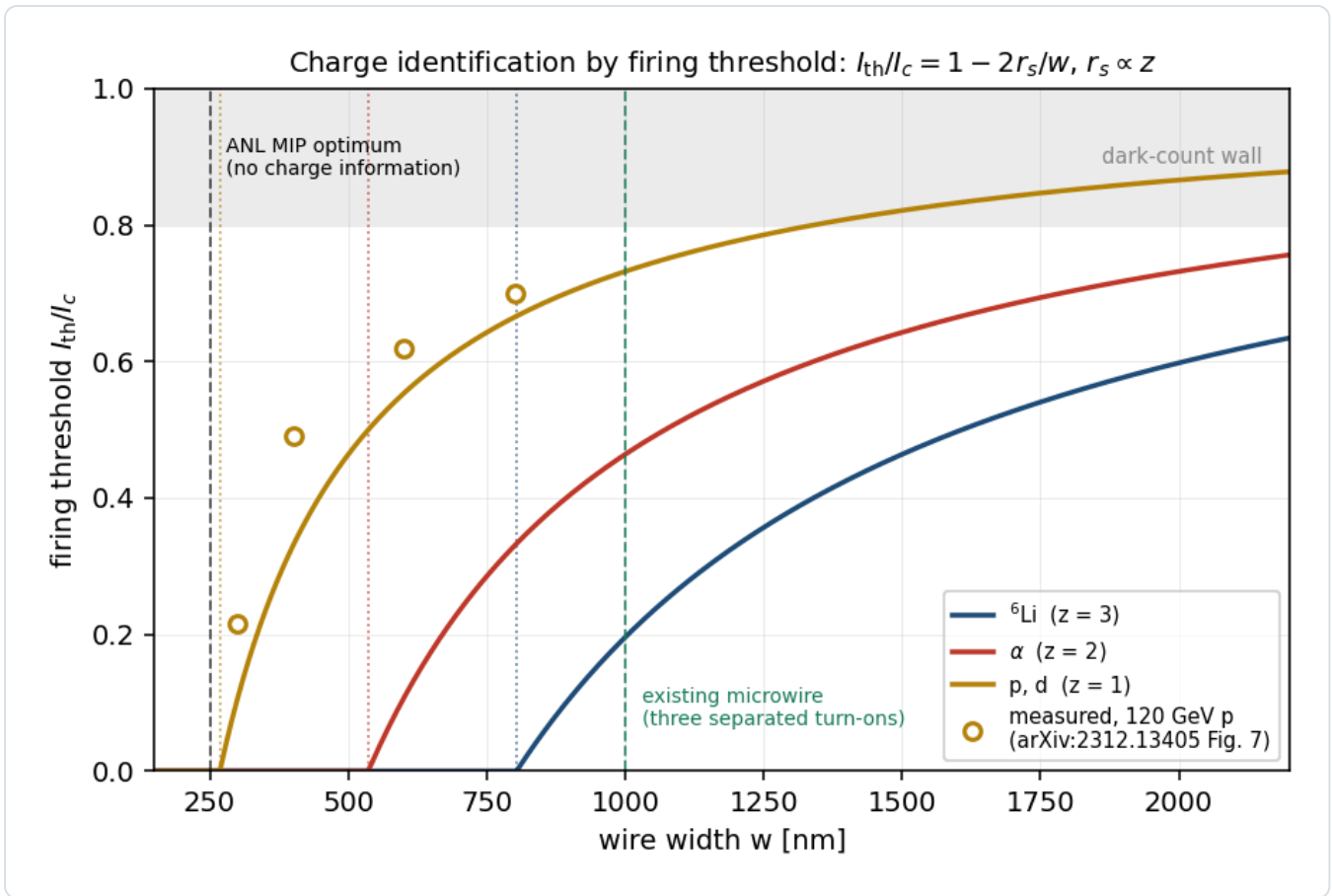


Figure 3 • Z by firing threshold. I_{th}/I_c against wire width for the three species, from $r_s \propto z$ anchored on Argonne's extrapolated 134 nm for a 120 GeV proton. Open circles are their measured thresholds versus wire width, digitised from Fig. 7 of [4]. Dotted verticals mark $w = 2r_s$ for each species — below that width the threshold is zero, the wire fires at any bias, and it carries no charge information at all. The measured points sit above the $z = 1$ curve because inverting each of them for r_s gives 102–118 nm (mean 113) against the 134 nm Argonne quote as their extrapolated "hard" hot spot; every design number here is a *ratio* between species, so the 15% offset moves nothing — re-anchoring on 113 nm leaves 1 μm separating all three below the noise wall (`evgen/tests/test_nearbeam.py`). `nearbeam_sensor_budget.py` .

WIRE WIDTH	I_{TH}/I_C , P/D	A	${}^6\text{Li}$	SEPARATED TURN-ONS
250 nm	0.00	0.00	0.00	none — ANL's MIP optimum
400 nm	0.33	0.00	0.00	none
800 nm	0.67	0.33	0.00	α from d
1000 nm	0.73	0.46	0.20	all three — the existing microwire
1500 nm	0.82	0.64	0.46	all three

At $w = 1000$ nm the three turn-ons sit at 0.20, 0.46 and 0.73 I_c . **Two bias points, at 0.33 and 0.60 I_c , tag Z by the firing pattern alone** — a $Z = 3$ plane that is blind to α and d, and a $Z \geq 2$ plane that is blind to d — and both sit well below the 0.80 I_c at which Argonne measure the dark-count rate rising exponentially [4]. The scheme is not speculative in kind: bias-point charge-state discrimination is a granted patent and was demonstrated on singly- versus doubly-charged lysozyme by subtracting two bias points [9].

Two things are in its favour. (i) The Z-ID operating point is also the blackbody-immune one. A 300 K surface radiates $\approx 3 \times 10^{18}$ photons $\text{cm}^{-2} \text{s}^{-1}$ above the 0.04 eV SNSPD threshold, so a photon-mode device cannot look at a warm beam pipe at all. Biasing at 0.33 and 0.60 I_c puts the wire below its photon threshold and leaves only the hard-hot-spot particle mode — which is what the scheme wants anyway. **(ii) The area answer and the Z-ID answer are the same device.** Argonne's efficiency optimum, $w \approx 250$ nm, is precisely $2r_s$ for a proton: maximal MIP efficiency and *zero* charge information. Z-ID needs deliberately *wide* wires — the microwire geometry Caltech/JPL/Fermilab already fabricate at 1 μm [7].

Three things are against it, and together they are decisive. (i) $r_s = 134\text{ nm}$ is not a measurement. It is the extrapolated zero-crossing of a four-point fit, and the authors write "While the physical validity of this simple model is a question of future work" [4]. Inverting their four points individually gives 102–118 nm (mean 113). Quote it as a model parameter, not a datum. (ii) **The volume in which Q is counted is unresolved by a factor of 5000.** The 2024 paper defines Q as "the energy that the proton has deposited into the thin film" — $\approx 20\text{ eV}$. The 2026 paper, with overlapping authorship, applies the same $\sqrt{Q}$ scaling to the substrate deposit — 0.1 MeV for the same proton [5]. Both scale as z^2 , so $r_s \propto z$ survives; the absolute radii do not. (iii) **"An interpolation, not an extrapolation" is only half true.** Argonne's ^{241}Am α differs from their 120 GeV proton almost entirely through $1/\beta^2$ ($\beta = 0.054$ against ≈ 1), not through z^2 . Their α point tests the $\sqrt{Q}$ law across a wide energy range; it says nothing about z^2 scaling at fixed velocity. **Nobody has ever varied Z at fixed β on one of these devices**, and every published ion demonstration — He^+ , lysozyme, the protein beams, the Ar ions — varies kinetic energy through an acceleration voltage, so charge enters as $E = zeV$, linear in z rather than z^2 from dE/dx .

4.1 • The incumbent carries more information — but far less decisively than a σ suggests

EICROC provides, per channel, a slow shaper path into an **8-bit 40 MHz successive-approximation ADC** for charge alongside its 10-bit 25 ps time-of-arrival TDC — the ToT of ALTIROC was replaced by that ADC specifically to handle dynamic range. Behind it sits an AC-LGAD with a **30 μm active thickness** at 500 μm pitch [11].

A σ is the wrong figure of merit, and this note said so with one. Gap over the quadrature sum of two Landau core widths is neither a PID separation power nor a fake rate. What puts an α inside a ^6Li 's acceptance window is the Landau *upper tail*, which falls as $1/\lambda$ — nothing like a Gaussian. Restated as a fake rate at a matched efficiency, with the Landau sampled rather than approximated (`nearbeam_zid_power.py` , 4 planes, 95% ^6Li efficiency, 1.5×10^6 events, 6.7×10^{-7} statistical floor):

READOUT	A FAKE RATE	EFFICIENCY REACHED
8-bit charge, per-plane likelihood ratio — the Neyman-Pearson optimum	2.3×10^{-5}	0.950
one bit per plane , majority-of- k , 99% efficient planes	3.1×10^{-5}	0.950
truncated mean — the standard analogue dE/dx estimator	2.7×10^{-3}	0.950
plain sum of the four planes	5.3×10^{-2}	0.950

One bit costs a factor 1.4, not orders of magnitude. The species are far apart (MPV 31.7 against 75.2 keV) and the discriminating power comes from requiring *coincidence* across planes, not from precision within one. Note also that a truncated mean is **86 $\times$ worse than one bit**, and a plain sum worse still: a Landau has no mean, so a single δ ray drags it. More bits does not automatically mean better Z-ID.

4.1a · Where the nanowire actually loses: fill factor

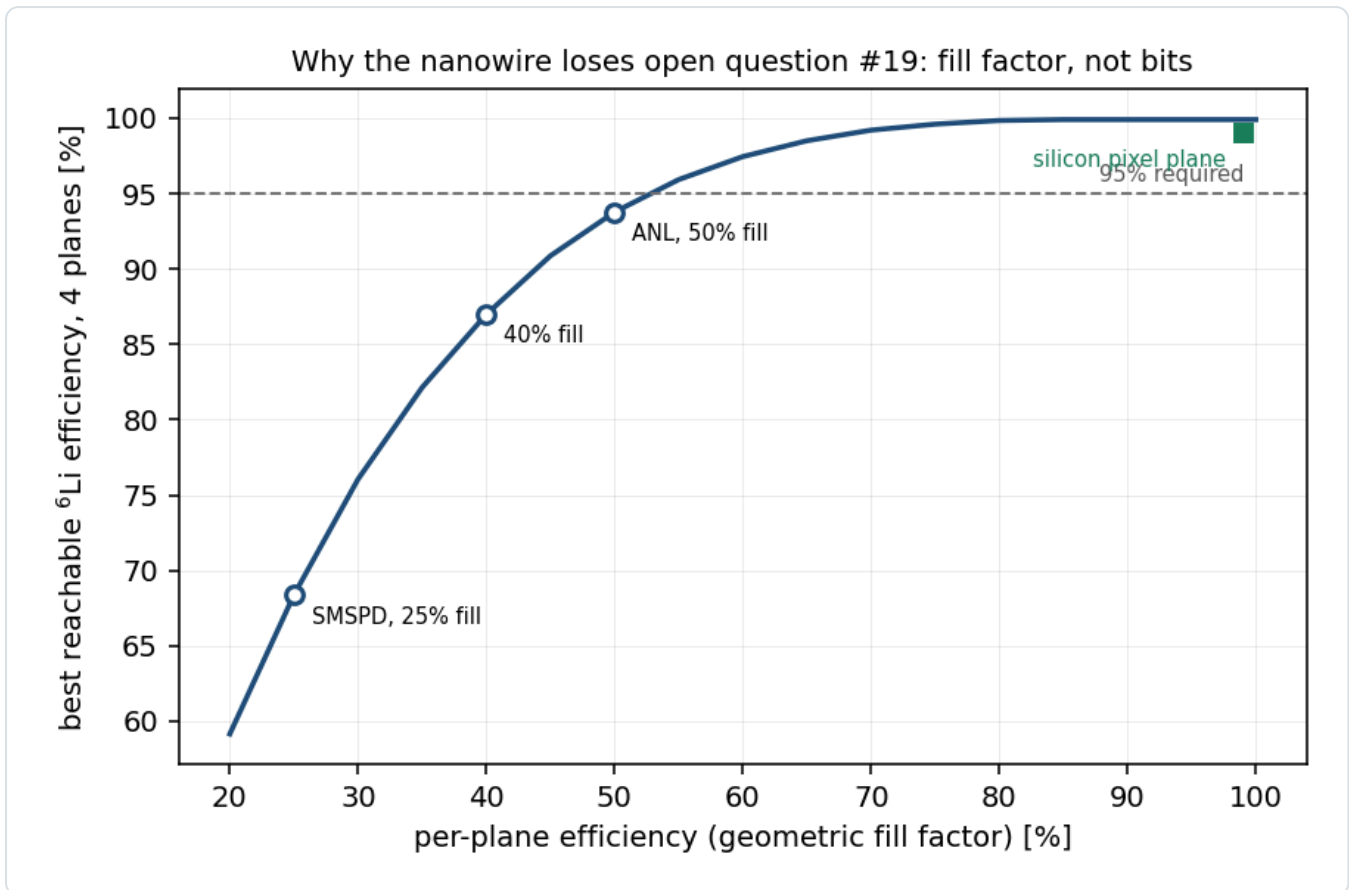


Figure 4 · Fill factor, not bits. The coincidence that makes one bit sufficient needs every plane to record the track. A silicon pixel plane does so ~99% of the time; a superconducting wire comb only over its geometric fill — 25% (SMSPD, [7]), 40% ([8]), 50% (the ANL EIC-targeted device, [4]). At 50% a four-plane array tops out at 93.7% ${}^6\text{Li}$ efficiency and *cannot reach 95% at any working point*. `nearbeam_zid_power.py`.

This is a **fabrication** number, not an information-theoretic one — which makes it a fairer and more actionable thing to put to the MEP group than "you only have one bit", and it is the one item on this list they could straightforwardly change.

The unknown that outranks all of it. EICROC's 8-bit full-scale range is not published anywhere we could reach. A ${}^6\text{Li}$ deposits $75\text{ keV} = 3.34\text{ fC}$ unamplified, **$\approx 67\text{ fC}$ at LGAD gain 20** (α : 28 fC), while EICROC0 is documented as "sensitive to low charge (2 fC)" and characterised at 12.5 fC. *If the pot front end is MIP-ranged, the ${}^6\text{Li}$ clips, α and ${}^6\text{Li}$ pile up at full scale, and the incumbent's Z-ID — the load-bearing rejection in this whole programme — is gone.* Compounding it, LGAD gain suppression for highly ionising particles is measured and reaches ~ 2.5 , compressing α and ${}^6\text{Li}$ toward each other in collected charge. That is §7 D4, and it should outrank every sensor comparison in this note.

4.2 · The honest limit on any dE/dx scheme

The limit is the **δ -ray upper tail**, and it is the same for every technology: it falls as $1/\lambda$ and is therefore nearly independent of sample thickness. Thickening does not rescue it; independent planes do, because an α needs four upward fluctuations rather than one. That is why the four-plane numbers above sit at 10^{-5} while any single plane is at the percent level — and why the *estimator* matters more than the bit depth.

One caveat runs the other way and makes every fake rate here conservative: the Landau is sampled untruncated, but a δ ray above $\sim 50\text{ keV}$ escapes a $30\text{ }\mu\text{m}$ layer and never deposits. The more tail-sensitive the estimator, the more its quoted fake rate is an overestimate.

4.3 · The handle that is better than either, and free

The background #19 exists to reject is ${}^6\text{Li} \rightarrow \alpha + d$. Both fragments sit at beam rigidity, so neither is separated by dispersion — but the breakup's relative momentum is *transverse* and is therefore not boosted. The α at $4p_u$ and the d at $2p_u$

take opposite kicks of the same k_T .

Two momentum scales, and this note first mixed them. $\sqrt{2\mu Q}$ with $\mu = 1247.7$ MeV gives **60.7 MeV/c** for $Q =$ the $\alpha+d$ separation energy 1.4743 MeV — the *bound-state* scale κ , which is what `spectator.sample_k` samples — and **42.2 MeV/c** for $Q = 0.712$ MeV, the decay momentum of the 2.186 MeV 3^+ resonance. Both are physical, for different production mechanisms. The table is now quoted from the repository's own density rather than a single k ; the earlier 6.7 / 18.4 / 44.8 mm used $k = 40$ MeV/c and were $\approx 1.6\times$ low.

OPTICS	MEDIAN SEPARATION	16–84%	IN 500 MM PIXELS
18 × 275	10.9 mm	4.9–22.7	22
10 × 100	30.1 mm	13.4–62.3	60
5 × 41	73.4 mm	32.6–152	147

$R_{12} = 30.6$ m, measured at 18 × 275; the other rows carry that lever arm for want of the per-optics value (§7 D3), so read them as scaling.

But the merge is not the dominant one-hit topology. At 5 × 41 the median separation exceeds the pot half-height, so the commoner failure is the partner flying *past the outer edge* or into the beam hole: single-fragment loss is of order **0.9 / 8.8 / 27% per breakup**, against a merge probability bounded by 0.26 / 0.035 / 0.006%. Neither pitch nor momentum resolution recovers a fragment that was never on the sensor. A merged pair, when it happens, deposits $\Sigma z^2 = 5$ against 9 and the charge readout separates it at $\approx 4\sigma$ per plane wherever it sits.

An intact ${}^6\text{Li}$ is one hit; the breakup is two, tens of pixels apart, in sensors that already exist. Topology is a stronger discriminant than dE/dx for exactly the background #19 was written about, and nobody in this programme had looked at it.

5 • What stands in the way

None of the following is a physics objection. All of them are engineering or programmatic, and all of them are load-bearing.

OBSTACLE	THE NUMBER	STATUS
Cold-stage power budget	A NbN device needs 2.8–3 K, a WSi microwire 0.8 K. SNSPD cold-stage loads are hundreds of μW to a few mW. A 4 K pulse tube delivers 1.5 W at 4.2 K and <i>zero</i> at its 2.8 K no-load floor, on a 190 kg water-cooled compressor drawing 10.7 kW.	hard
Beam-induced RF heating	A ferrite-shielded TOTEM Roman Pot dissipates ≈ 40 W at the LHC. Scaling the resistive-wall/broadband factor $Q^2 M / \sigma_z^{3/2}$ to EIC parameters gives $\approx 16 / 8 / 0.8$ W at highest-ECM / highest-L / 41 GeV. Even the 41 GeV case exceeds a pulse tube's entire 4.2 K fill.	hard
In-vacuum precedent	None found anywhere for a biased superconducting detector inside an accelerator's <i>beam</i> vacuum. The precedents — LHC CryoBLM silicon and diamond at 1.9 K, the CERN AD SQUID current comparator — sit outside it, and the AD beam is $\sim 10^7$ antiprotons against the EIC's 8×10^{13} protons.	open
Vibration	Cryocooler cold-head displacements are specified at 7–9 μm ; a purpose-built decoupling structure brings that to 10 nm but is heavy and stiff — the opposite of a pot that rides motorised rails to within 10σ of the beam every fill.	mitigable
Radiation hardness	Unmeasured. The FY22 proposal's central deliverable was <i>"an upper limit for the accumulated dose ... This currently unknown limit"</i> ; four years on, no publication reports the LEAF irradiation, the JLab Hall C parasitic run or the SEU cross-section. The 2023 Quantum Sensors for HEP report says SNSPDs <i>"are expected to be radiation hard ... Although published studies in this area are lacking"</i> . The whole "radiation hard, therefore closer" argument is an anticipation — the proposal itself writes "(anticipated)".	unmeasured
Area	The largest SNSPD/SMSPD ever run in a GeV particle beam is 2 × 2 mm ² with 8 channels [8]. A near-beam strip 3 mm deep over 20 mm of vertical extent, both sides of the beam, is 120 mm ² — 30× the state of the art, and 1000× if the whole 9.3 mm gap is instrumented. Tiled as 30 × 30 μm^2 nanowire devices that is 133,000 channels; as 1 × 1 mm ² microwire tiles it is 960.	funded R&D

Latching	Latching is a hard failure, not a dead time: the device locks resistive and stops detecting. Argonne's ²⁴¹ Am α run is the reassurance — a $\approx 1\text{ }\mu\text{m}$ hot spot did <i>not</i> latch a 100–200 nm wire, and gave clean count-rate plateaus [5].	addressed
Rate	Not a blocker. The ePIC Preliminary Design Report puts pot rates at a few Hz per channel in normal running and "30–50 kHz at $\sim 10\sigma$ " from beam halo, on STAR experience — four orders below the $\sim 10^8$ counts/s a nanowire's fall time allows [6].	closed

Cryogenics also inverts the packaging argument. The nanowire's advantages — windowless operation in the beam vacuum, an active edge a few hundred nanometres from the substrate rim, no vacuum-window material — are real and are exactly what a near-beam sensor wants. But a device that cannot see a 300 K surface must be enclosed in a cold shield, and a cold shield is material in front of the sensor. The windowless argument and the blackbody constraint pull against each other, and the resolution — bias below the photon threshold, §4(ii) — is available but has never been demonstrated next to a beam.

6 · A baseline correction: the ePIC pot geometry has moved

Independent of nanowires, and more immediately actionable. The aperture this repository measured was taken in the September-2024 epic-main inside jug_xl-nightly. Reading the current main branch of eic/epic directly [3]:

	SEPTEMBER-2024 (WHAT TOOLS/FULLSIM MEASURED)	CURRENT MAIN
module	32 × 32 mm	16 × 16 mm
insertion	energy-independent: 3.2 cm outer, +0.7 cm central	per-energy 10 σ offsets in beamline_*.xml
RF shield	1 mm aluminium on each module face, active	commented out — "we don't know if we will even need it ... Oct. 2025"
implied blind block at 18 × 275	32 mm (x) × 7 mm (y)	16 mm (x) × 2.7 mm (y)

The old geometry's 32 mm block gives 32 mm / 30.6 m = 1.046 mrad against the 1.03 mrad the scan measured — agreement to 1.5%, which is the check that the measurement and the file reading confirm each other.

The correction runs in opposite directions at the two ends, which is why it cannot be guessed. At 18 × 275 the current 16 mm block implies ≈ 0.52 mrad — below the 0.727 mrad 10 σ envelope, so the sensor package no longer binds and the beam does. At 5 × 41 the per-energy insertion moves the other way, to a 29.6 mm inner edge, because the pots now retract properly for the larger low-energy beam — so the aperture there gets worse, not better. That is precisely the configuration at which the coherent programme was said to survive.

What this means for the numbers in circulation. The measured aperture, the sign-flipped $\langle \cos 2\beta \rangle = -0.77$, the 1.4×10^{-2} tagged fraction, and the conclusion that "the coherent programme is a low-energy programme" are all conditioned on a superseded geometry. They should be re-measured against the current release before being quoted further. The conversion from angle to millimetres also needs the per-optics R_{12} , which was measured for 18 × 275 alone (30.6 m); this note therefore works in angle throughout and quotes millimetres only for that configuration. **The curves of Figure 1 are unaffected** — they price every aperture, which is why they are plotted that way; what moves is only the marker labelled "you are here".

7 · Can the tag be recovered by tuning the beam and placing detectors?

Asked properly, and answered by a dedicated feasibility study (2026-08-27). **The answer is no as this programme is currently projected, and partly-yes only under four conditions that are all open.** The corrected baseline is worse than §2 suggests: at the γ -matched energies and the machine's own per-configuration divergence the coherent tagged fraction is 8.4×10^{-8} at 5 × 41, and every lever below multiplied together buys back $\approx 10^5$ against a shortfall of 5×10^6 .

LEVER	GAIN AT 5 × 41	COST	CONFIDENCE
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Pot standoff 10σ → 8σ (active edge)	×216	no luminosity; halo rate and an aperture hierarchy that does not exist yet	medium — RHIC ran pot 8σ / <i>active edge</i> 10σ; an 8σ active edge is 2σ beyond anything demonstrated
Two-ring β* de-squeeze , factor F	×1.6×10 ⁵ at F = 3 ×1.8×10 ⁶ at F = 25	$L \propto 1/F$, and an equal ESR de-squeeze — a hadron-only raise multiplies ξ _p by F against a limit already at 0.015	low above F ≈ 10
Shrink the 16 mm blind block	×1 today; ×4.2 once F ≥ 5	a layout change already on ePIC's roadmap	medium , premise unresolved
Detector relocation along the FF line	≤ ×1.75, provably	—	negligible
Lower beam energy	×1 — already exhausted	—	γ = 43.5 is the lowest the machine can collide at
Colder beam from low intensity	negative	—	drop it — see below
Dispersive (rigidity) tagging at IP6	unavailable	—	drop it

Three levers that look promising and are not. (i) The cold-beam argument is backwards. HSR injection is at fixed *rigidity*, so an A/Z = 2 ion injects at roughly half the proton's γ and carries ≈2.5× *more* space charge; and EPIOS p. 13 says store cooling is a later-stage addition, so there is no cooling/IBS equilibrium to settle into at all. **(ii) Dispersive tagging at IP6 is 30–60× short.** The minimum detectable Δ(Bp)/(Bp) for a p_T ≈ 0 fragment is 0.1 (Gamage *et al.*, arXiv:2105.13564, Table 1) against a coherent signal at δ ≤ 1.7×10^{−3} — and YR §8.5.7 says "*it is not possible to instrument the beam pipe further downstream than around 30 m in the current EIC design due to the crab cavities*". **(iii) The LHC 3σ precedent does not transfer.** TOTEM/ALFA's "3σ" is the pot *window*; the silicon sat 4.8–5.4σ out, and those fills ran at ~3×10¹¹ protons total. The EIC's 10σ is the sensor's inner edge. The right precedent is RHIC, in this tunnel: β* = 22 m with Roman Pots in four dedicated stores in 2009, and a pot approach no closer than 8σ.

The one column that closes is not 5 × 41. At the top rigidity mode against a 1160-bunch 275 GeV electron fill, with F = 25 and an 8σ standoff, the projection reaches δa_e ≈ 8×10^{−4} — a 12σ measurement of a 1% amplitude. Four conditions gate it and all four are open: a ×25 two-ring de-squeeze; an 8σ active-edge standoff; proton-equivalent *lithium* luminosity in a 1160-bunch fill, which nobody has published; and ⁶Li polarization survived to γ = 147 with a species for which Siberian snakes do not work. At 5 × 41, even F = 25 with 8σ gives 1.4σ — the configuration is luminosity-starved at 0.44×10³³.

And the energy ordering should not be bet on. At published optics 5 × 41 beats the top mode by five orders; the ordering crosses over near F ≈ 4–5, exactly where the evidence runs out, and it is carried entirely by *proton* luminosities applied to a lithium beam. With a species-limited luminosity the advantage vanishes.

The two questions for C-AD, in one email: **can IR6 be matched to a hadron β* raised ×3, ×10 and ×25 in both planes — with the ESR co-de-squeezed so ξ_p stays ≤ 0.015 — and what are the resulting IP → pot phase advances?** And **what physically sets the 10σ retraction: the collimation hierarchy, halo lifetime, or convention?** The published CDR high-acceptance β* factors shrink monotonically toward low energy — ×5.21 at 18 × 275, ×1.49 at 10 × 100, and **×1.00 at 5 × 41**, where Tables 3.3 and 3.4 are identical — so the headroom the designers found runs out exactly where this programme needs it.

8 • Conclusion

There is no design here in which superconducting nanowires measurably improve far-forward lithium tagging, and both candidate roles fail on arithmetic rather than on unknowns. The near-beam role fails because the sensor's dead edge is worth 0.005 mrad against a gap of 0.5–3 mrad set by module tiling and the 10σ rule, and because the 10σ rule

protects the machine rather than the sensor. The charge-ID role fails because the incumbent AC-LGAD already digitises an 8-bit analogue charge over a 30 μm active layer — strictly more information per plane than a nanowire's one bit — across four planes ePIC already has, and because the limit on Z separation is the δ -ray tail, which is the same for both.

That is a conclusion about *this* application, not about the technology. The Argonne programme's own four-application list points elsewhere — the cold bore of a superconducting magnet, in front of the ZDC, a Compton polarimeter — and the field result that makes the technology unique (saturated efficiency to 5 T parallel to the device plane [10]) is worth nothing at a pot in a field-free drift and everything inside a magnet.

What this study leaves behind is worth more than its negative answer. Figure 1 prices any closer approach by any means, and says the mid-energy configuration is worth $\times 569$ and the top energy nothing. §6 says the geometry those numbers were anchored to no longer exists. And §4.3 says the background that motivated open question #19 is separable by hit topology in sensors that are already baselined. The ranked list of what would actually help the lithium tags is: the low-energy configuration; re-tiling the near-beam silicon, which ePIC is already designing; using the charge EICROC already digitises; two-hit topology for the $\alpha + d$ channel; and the IR-8 secondary focus. Superconducting nanowires appear nowhere on it.

References

- [1] R. Abdul Khalek et al., *Science Requirements and Detector Concepts for the Electron-Ion Collider: EIC Yellow Report*, Nucl. Phys. A 1026 (2022) 122447, arXiv:2103.05419, §14.5.3 (refs/2103.05419_part4.pdf): the four EIC SNSPD applications, of which the first is the far-forward Roman Pot; edgeless operation "within a few 100 nm of the substrate edge"; membranes "as thin as few 10 μm "; the mm^2 pixel-array goal.
- [2] W. Armstrong (PI) et al., *Superconducting Nanowire Detectors for the EIC*, FY22 EIC-Related Generic Detector R&D proposal #18, §3.3 Concept 1 — reviewed and partially funded (Report of the 2022 EIC-related Generic Detector R&D Review Committee, June 2023). The 10σ rule-of-thumb passage quoted in §3.
- [3] eic/epic , branch main : compact/far_forward/roman_pots_eRD24_design.xml and compact/fields/beamline_{18x275,10x100,5x41}.xml , read directly (2026-08-26). Module 1.6×1.6 cm; per-energy offsets with the comment "These are the ten-sigma cuts for the Roman pots, translated to the physical layout we currently have. They are not perfectly ten-sigma for reasons of physical geometry."; RF shield module components commented out, "Oct. 2025".
- [4] S. Lee, T. Polakovic, W. Armstrong, A. Dibos, T. Draher, N. Pastika, Z.-E. Meziani, V. Novosad, *Beam Tests of SNSPDs with 120 GeV Protons*, Nucl. Instrum. Meth. A 1069 (2024) 169956, arXiv:2312.13405 (refs/2312.13405.pdf). 12 nm NbN, $30 \times 30 \mu\text{m}^2$, wire widths 300–800 nm at 2.82 K; Eqs. 1–2; $r_s = 134$ nm; ~ 250 nm optimum; background rising exponentially above $I_b/I_c = 0.8$.
- [5] S. Lee, W. Armstrong, J. DiPreta, C. Dulya, V. Novosad, T. Polakovic, *Optimization of Cryogenic Detector Test Station by Rejecting Electromagnetic Interference*, Nucl. Instrum. Meth. A 1093 (2027) 171953, arXiv:2601.03158 (refs/2601.03158.pdf). The 5.5 MeV ^{241}Am α at $\approx 1 \mu\text{m}$ hot spot by the same $\sqrt{Q}$ scaling; clean plateaus with no latching; "a detailed study of α detection is underway".
- [6] ePIC Collaboration, *Preliminary Design Report*, 29 September 2024, the Roman Pots and off-momentum detectors section: "Rates during normal operations, with expected vacuum of 10^{-9} mbar, are a few Hz/channel. However, the beam halo could potentially provide rates of 30–50 kHz at $\sim 10\sigma$ from experience of Roman pots at STAR."
- [7] C. Wang et al. (Caltech / JPL / Fermilab), *Towards High-Efficiency Particle Detection Using Superconducting Microwire Arrays*, arXiv:2510.11725 (refs/2510.11725.pdf). 8-channel $1 \times 1 \text{ mm}^2$ WSi, 4.7 nm film, 1 μm wires on a 3 μm gap (25% fill), 0.8 K; 75% fill-factor-normalised efficiency and 130 ± 17 ps at CERN SPS H6.
- [8] C. Peña, C. Wang, S. Xie et al., *Characterization of a Superconducting Microwire Single-Photon Detector Array for Charged-Particle Detection*, JINST 20 (2025) P03001, arXiv:2410.00251 (refs/2410.00251.pdf). $2 \times 2 \text{ mm}^2$, 8 channels, $30 \pm 1 \mu\text{m}$ spatial resolution; amplitude distributions indistinguishable across species; sensor edge sharper than the 10 μm telescope.
- [9] R. Cristiano, M. Ejrnaes, A. Casaburi, N. Zen, M. Ohkubo, *Superconducting nano-strip particle detectors*, Supercond. Sci. Technol. 28 (2015) 124004: the hot-spot threshold in the ion literature; the charge-state discrimination demonstrated on lysozyme by bias-point subtraction (Suzuki et al., Rapid Commun. Mass Spectrom. 24 (2010) 3290; US Patent 8,872,109 B2, Ohkubo & Suzuki, AIST). Also J. J. Renema et al., arXiv:1301.3337, for the photon-side form of the same relation.
- [10] T. Polakovic, W. R. Armstrong, V. Yefremenko, J. E. Pearson, K. Hafidi, G. Karapetrov, Z.-E. Meziani, V. Novosad, *Superconducting nanowires as high-rate photon detectors in strong magnetic fields*, Nucl. Instrum. Meth. A 959 (2020) 163543, arXiv:1907.13059 (refs/1907.13059.pdf). Saturated efficiency to 5 T parallel to the device plane but only ≈ 0.5 T perpendicular; 10^7 counts/s measured, $\sim 10^8$ /s from $\tau_F = 11.78$ ns.

Appendix A · Revision history

DATE	REVISION
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2026-08-27 (later)	§7 replaced by the outcome of a dedicated recovery study. The corrected baseline at the γ-matched energies is 8.4×10^{-8} at 5×41, and the levers together buy $\approx 10^5$ against a 5×10^6 shortfall. Three levers that looked promising are dropped with reasons: the cold-beam argument is <i>backwards</i> (fixed-rigidity injection puts an $A/Z = 2$ ion at half the proton's γ with $\approx 2.5\times$ more space charge, and EPIOS says store cooling is later-stage, so there is no equilibrium to reach); dispersive tagging at IP6 is 30–60$\times$ short of the 0.1 rigidity threshold and the beam pipe cannot be instrumented past ~ 30 m because of the crab cavities; and the LHC 3σ precedent is a pot <i>window</i>, not an active edge — the right precedent is RHIC's $\beta^* = 22$ m stores in this tunnel, which approached no closer than 8σ. The β^* lever survives but is a two-ring change, since a hadron-only de-squeeze multiplies ξ_p by the same factor.
2026-08-27	The charge argument is corrected, and it was overstated. "4.8σ per plane" is neither a PID separation power nor a fake rate; restated as a sampled-Landau fake rate at matched efficiency, one bit per plane costs a factor 1.4 against the Neyman-Pearson optimum, and a truncated mean costs a factor 86 to one bit (§4.1, Figure 4, <code>nearbeam_zid_power.py</code>). The nanowire loses on geometric fill factor — 50% fill cannot reach 95% ^6Li efficiency over four planes — not on bit depth. Also corrected: the α+d separation mixed the bound-state scale $\kappa = 60.7$ MeV/c with the 3^+ resonance decay momentum 42.2 MeV/c, and is now quoted from the repository's own density (median 10.9 / 30.1 / 73.4 mm, $\approx 1.6\times$ the earlier single-k figures, §4.3); the dominant one-hit topology is single-fragment <i>loss</i> (0.9 / 8.8 / 27% per breakup), not merging ($\leq 0.26\%$); and EICROC's unpublished dynamic range is promoted to the top of §7.
2026-08-26	First version, then revised the same day against an adversarial review. The verdict changed sign. As first written, this note concluded that charge identification was the strong half of the case; the review established that the incumbent AC-LGAD already digitises an 8-bit charge over a 30 μm active layer across four planes, giving $\approx 4.8\sigma$ of $\alpha \leftrightarrow ^6\text{Li}$ separation per plane against a nanowire's one bit (§4.1) — so a nanowire is a downgrade, not an upgrade, and open question #19 was being asked of the wrong technology. Three further corrections: $r_s = 134$ nm is an <i>extrapolated</i> model parameter and not a measurement; the volume in which Q is counted differs by a factor 5000 between Argonne's own two papers; and their ^{241}Am α point tests the $\sqrt{Q}$ law across β, not z^2 at fixed velocity, so the "interpolation" argument is only half true (§4). The dead-edge argument of §3 was retired as worth 0.005 mrad against a 0.5–3 mrad gap, the granularity argument redirected to the x-moving layer ePIC is already designing, and §4.3 added — the $\alpha + d$ breakup separates by 13 to 90 pixels of the existing 500 μm pitch, which is a better answer to #19 than any dE/dx scheme and is free. The original content: The aperture scan and the coherent chain at both apertures (Figures 1–2, <code>nearbeam_aperture_scan.py</code>, <code>nearbeam_reach_gain.py</code>, and <code>--near-beam-mrad</code> added to <code>money_cos2phi_coherent_reco.py</code>); the hot-spot firing-threshold model of charge identification (Figure 3, <code>nearbeam_sensor_budget.py</code>); the obstacle table; the ePIC geometry correction of §6, read from the current <code>main</code> branch; the question list of §7.

All acceptances use $B = 50 \text{ GeV}^{-2}$ for the coherent $|t|$ slope and the high-acceptance optics ($\sigma_\theta = 72.7 \mu\text{rad}$, $10\sigma = 0.727 \text{ mrad}$). The measured silicon aperture is from `tools/fullsim` in the September-2024 `epic-main` — see §6. Energy deposits are Bethe means in a 12 nm NbN film; the Landau most-probable value is deliberately not quoted, because $\xi/l \approx 0.002$ in this film puts it far outside the $\xi \gg l$ regime the formalism requires. Hot-spot radii are anchored on Argonne's *extrapolated* 134 nm — a fit parameter of a model whose authors write that "the physical validity of this simple model is a question of future work" — and scaled as z . Built by `reports/build_report.py`; figures from the `evgen/scripts` named in the text; display mathematics typeset at build time.